

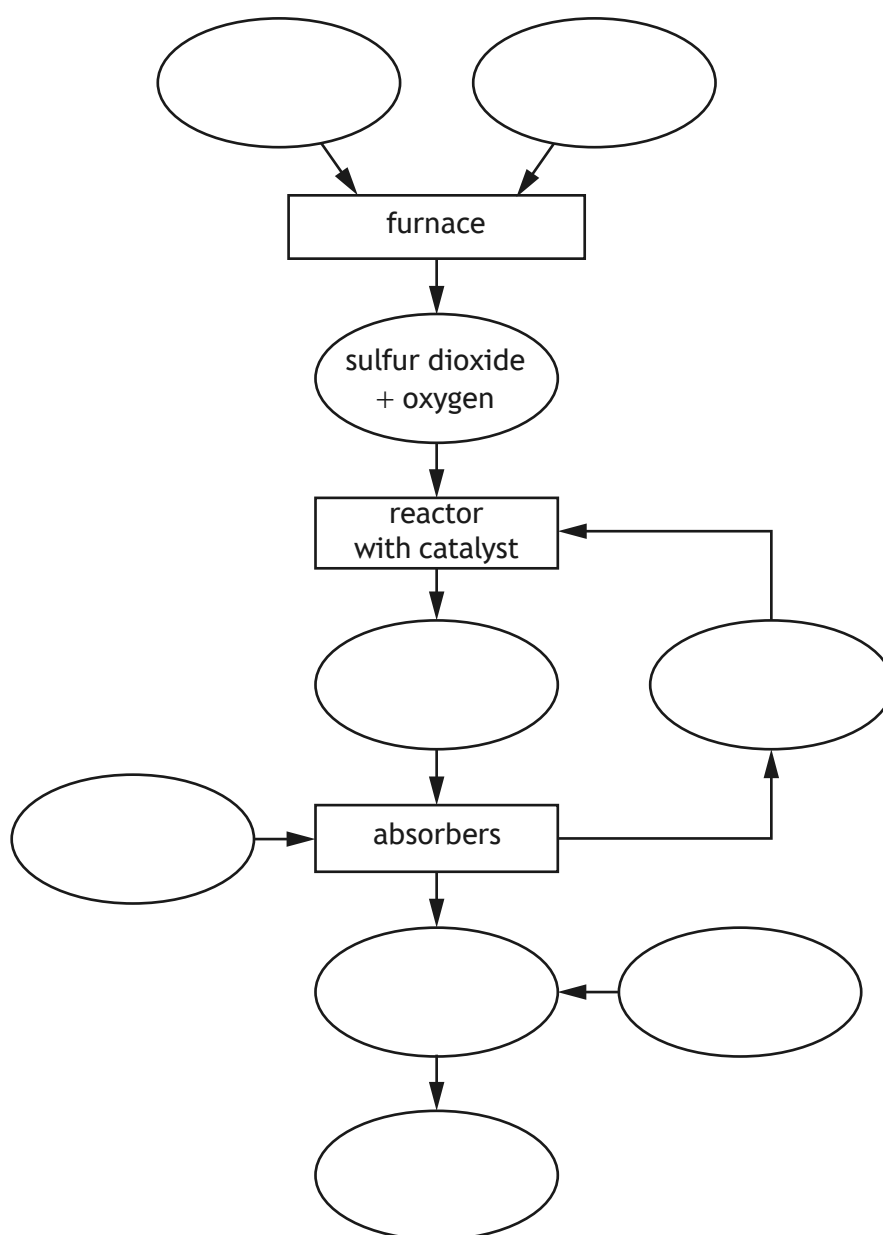
7. Sulfuric acid is an important chemical with many uses in industry. The main process used to make sulfuric acid is the Contact Process.

- (a) The Contact Process starts when sulfur is burned in a furnace with excess air. This forms a gas mixture that contains sulfur dioxide and oxygen. The gas mixture is cooled and then passed into the reactor which contains a catalyst. Sulfur trioxide is formed and then passed into absorbers where it is absorbed into concentrated sulfuric acid to form a product called oleum. The oleum is diluted with water to give sulfuric acid. Unreacted sulfur dioxide is not absorbed and is recycled back into the reactor.

Use the information above to complete the flow diagram for the Contact Process.

2

(An additional diagram, if required, can be found on *page 39*.)



7. (continued)

- (b) The reactions involved in the Contact Process are highly exothermic.

State a disadvantage of industrial processes that involve reactions that are highly exothermic.

1

- (c) Sulfur and oxygen contain London dispersion forces.

(i) Explain how London dispersion forces arise.

2

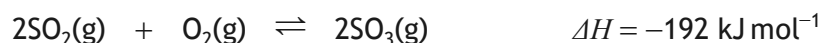
(ii) Explain fully why the London dispersion forces in sulfur are stronger than those in oxygen.

2



7. (continued)

(d) The reaction that occurs in the reactor of the Contact Process is



Circle the correct statement in each column of the table to show the effect of using a catalyst in the reaction.

1

Effect of catalyst on enthalpy change	Effect of catalyst on activation energy
increase stay the same decrease	increase stay the same decrease

(e) One use of sulfuric acid is in the production of soapless detergents.

(i) State the advantage that soapless detergents have over soap when used with hard water.

1

(ii) Describe the key structural features of a soapless detergent molecule.

1

[Turn over



* X 8 1 3 7 6 0 1 2 1 *